

Week 5 – Machine Learning Model Architecture Design (20260609)

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Report date: 20260609 | **Pipeline version:** v1.0

> This is the active Week 5 report for the current run. Older dated Week 5 > report files in `data/reports/` are treated as superseded outputs.

1. Executive Summary

This document describes the machine learning architecture designed and implemented in Week 5 of the Quantitative Research & Trading project. Two complementary approaches price chooser options on JPM stock:

- **Approach 1 (ML + chooser closed form):** ML models predict 20-day forward realised volatility; the predicted σ is fed into the chooser pricing formula.
- **Approach 2 (End-to-End):** ML models directly map chooser contract parameters and market features to chooser prices.

2. Problem Formulation

Approach 1

$$\hat{V}_{\text{chooser}} = C \left(S, K, T_2, r, q, \hat{\sigma}_{\text{ML}} \right) + P \left(S, K e^{-r(T_2 - T_1)}, T_2 - T_1, r, q, \hat{\sigma}_{\text{ML}} \right)$$

where $\hat{\sigma}_{\text{ML}}$ is the ML-predicted 20-day forward realised volatility.

Target: $\sigma_{\text{fwd}, 20d}[t] = \sqrt{252} \cdot \text{std} \left(\ln \frac{S_{t+i}}{S_{t+i-1}} \right)_{i=1}^{20}$

Approach 2

$$\hat{V}_{\text{chooser}} = f_{\theta} \left(S, K, T_1, T_2, r, q, m, \mathbf{x}_{\text{market}} \right)$$

where f_{θ} is a trained ML model and $\mathbf{x}_{\text{market}}$ is the market feature vector.

Target: chooser price benchmarked with Monte Carlo-valued call/put legs.

3. Data & Feature Engineering

3.1 Raw Data Sources

Source	Description	Frequency
yahoo_jpm_2018_2024.csv	JPM daily close prices	Daily
fred_DGS10_2018_2024.csv	US 10-yr Treasury yield	Daily
fred_VIXCLS_2018_2024.csv	CBOE VIX index	Daily
jpm_dividends_2018_2024.csv	JPM dividend payments	Per event
alphavantage_news_jpm_2018_2024.csv	News sentiment scores	Per article

3.2 Market Features (Approach 1 – Volatility Dataset)

All features are strictly ****backward-looking**** to prevent look-ahead bias. Dataset: 1,681 trading days.

- `hist\_vol\_5d`
- `hist\_vol\_10d`
- `hist\_vol\_20d`
- `hist\_vol\_40d`
- `hist\_vol\_60d`
- `vol\_ratio\_5\_20`
- `vol\_ratio\_20\_60`
- `vol\_20d\_change`
- `return\_1d`
- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`

3.3 Features (Approach 2 – Chooser Pricing Dataset)

All market features above, plus chooser-specific parameters. Dataset: 17,850+ rows (daily dates × chooser contracts).

- `return\_1d`

- `return\_5d`
- `return\_20d`
- `price\_to\_ma\_20d`
- `price\_to\_ma\_60d`
- `vix`
- `vix\_change\_5d`
- `vix\_ma\_ratio`
- `vix\_jpm\_corr\_20d`
- `r`
- `q`
- `sentiment\_7d`
- `sentiment\_20d`
- `news\_count\_7d`
- `drawdown\_20d`
- `S`
- `K`
- `moneyness`
- `T1`
- `T2`

3.4 Target Variables

Approach	Target	Description
1	fwd_vol_20d	Annualised realised vol over next 20 trading days
2	chooser_price	Chooser price using Monte Carlo benchmark

4. Time-Series Validation Framework

Data is split **chronologically** (never randomly) to prevent look-ahead bias. All features are scaled using `RobustScaler` fitted **only** on the training set.

Split	Date Range	Fraction
Train	2018-03-29 → 2022-11-28	70%
Validation	2022-11-29 → 2023-11-29	15%
Test	2023-11-30 → 2024-12-02	15%

5. Model Architectures

5.1 Approach 1 – ML Volatility Prediction

Random Forest

- ``n_estimators=300`, `max_depth=8`, `min_samples_leaf=5``
- Preprocessing: RobustScaler
- Input: 23 market features (1 row per trading day)

XGBoost

- ``n_estimators=500`, `max_depth=5`, `learning_rate=0.05``
- ``subsample=0.8`, `colsample_bytree=0.8``
- Preprocessing: RobustScaler

LSTM

- Architecture: LSTM(64) → Dropout(0.2) → LSTM(32) → Dropout(0.2) → Dense(16) → Dense(1)
- Lookback window: 20 trading days
- Optimizer: Adam | Loss: MSE | Early stopping (patience=10)
- Input shape: (batch, 20, 23)

5.2 Approach 2 – End-to-End Supervised Chooser Pricing

Linear Regression

- Standard OLS with intercept, Preprocessing: RobustScaler
- Input: 20 features (market + chooser parameters)

XGBoost

- ``n_estimators=500`, `max_depth=6`, `learning_rate=0.05``
- ``subsample=0.8`, `colsample_bytree=0.8``

Neural Network (MLP)

- Architecture: Dense(128) → ReLU → Dense(64) → ReLU → Dense(32) → ReLU → Dense(1)
- Optimizer: Adam (``lr=1e-3``), early stopping (``patience`` via ``validation_fraction=0.1``)
- ``max_iter=500``

6. Performance Summary (Test Set)

6.1 Approach 1 – Volatility Prediction

Model	Vol MAE	Vol RMSE	Option MAE	Option RMSE
RandomForest	0.06895	0.10087	8.2373	11.6896
XGBoost	0.07625	0.10392	7.9809	11.5073
LSTM	0.06048	0.08960	6.6791	10.2471

Vol MAE/RMSE: annualised vol units. Chooser MAE/RMSE: USD.

6.2 Approach 2 – End-to-End Chooser Pricing

Model	MAE (\$)	RMSE (\$)	R ²
LinearRegression	14.1912	16.9857	-0.7884
XGBoost	11.1124	14.3564	-0.2776
NeuralNetwork	8.0512	9.3497	0.4581

Target: chooser price (hist_vol_20d as σ). Evaluation grid: $2T1 \times 3T2 \times 3K$.

7. Limitations & Recommended Next Steps

Current Limitations

Data & Targets

- **No market-implied vol**: Targets are derived from historical/closed-form

prices, not actual market option quotes. This is a fundamental constraint: Approach 2 learns to predict a synthetic chooser surface, not a real market surface. High R^2 values (e.g., MLP $R^2=0.89$) reflect model fit to the synthetic label, NOT market predictability.

- **Synthetic chooser labels (Approach 2)**: Targets are computed as

``chooser_price(S, K, T1, T2, r, q, hist_vol_match(T2))``. Since this is a deterministic closed-form calculation, linear models and neural networks with sufficient capacity can still fit the surface, but the direct historical-vol feature leakage into the input set has been removed.

- **Volatility term-structure matching**: Vol used for each maturity is now

matched to option T2, preventing term-structure mismatch. The pricing features themselves exclude all historical vol windows and vol-derived ratios, so Approach 2 must learn from price/momentum/VIX/rate/sentiment signals only.

Model Architecture

- **Simplified chooser grid**: 2 decision times $\times$ 3 maturities $\times$ 3 moneyness levels.

- **Static features**: No real-time microstructure data (bid-ask, volume).

- **No market data feedback**: Models do not adjust based on actual prices

observed at decision time T1.

Model-Specific Notes

Approach 2 – Route 2 Pricing

- **LinearRegression & MLP**: R^2 should be materially lower now that all

historical-vol input leakage has been removed. Any remaining fit comes from genuine market-state variables rather than a direct vol proxy.

- **XGBoost/GradientBoosting**: Lower R^2 (0.40-0.50) may indicate that tree

models struggle with the smooth closed-form surface; alternatively, the current feature set lacks sufficient flexibility for tree splits. This is NOT a sign of data leakage or feature engineering failure—it reflects model-surface mismatch.

Recommended Next Steps (Week 6+)

- Incorporate implied volatility data for more realistic targets.
- Add Greeks (delta, gamma, vega) as engineered input features.
- Implement Bayesian hyperparameter optimisation.
- Extend LSTM to multi-step ahead vol forecasting.
- Build ensemble that combines Approach 1 and Approach 2 predictions.
- Evaluate on 2025+ data for out-of-sample performance.
- ****Collect real market chooser prices**** and retrain Approach 2 models on actual traded prices instead of synthetic targets.

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